HSAN ELLOUZE

Software Engineering Student | Software Engineer | Full-Stack, AI & Cloud
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Seeking a 6-month end-of-studies internship (PFE) from February 2027 | Open to relocation

Summary

Final-year Software Engineering student at INSAT with experience building and deploying production web applications and AI-powered systems. Combines full-stack and backend development (Python, TypeScript, REST APIs) with RAG/LLM integration, databases, testing, Docker, CI/CD, and observability projects. Comfortable contributing across the software lifecycle, from system design and implementation to deployment and troubleshooting.

Experience

AI Engineer Intern – Applicare.ai / Sahera, Tunis Jun 2026 – Aug 2026
- Built AskSahera, a bilingual (French/English) medical RAG assistant on perimenopause and auriculotherapy, grounding answers in scientific literature.
- Implemented ingestion for PubMed XML and PDFs using GROBID/TEI-XML; normalized article metadata, applied IMRAD-aware semantic chunking with sliding-window fallback, and indexed chunks in Qdrant.
- Integrated Claude for grounded response generation and batched interpretation of extracted tables; migrated embeddings from OpenAI to Gemini to manage costs.
- Stored source PDFs in Cloudflare R2 and deployed the Express.js/Node.js API on Vercel, running GROBID separately in Docker with environment-based secret management.
- Built a Three.js/WebGL 3D ear model with 13 validated auriculotherapy points, synchronized bidirectionally with chat responses; contributed to a paper submitted to IEEE SIME 2026.

Mazdev – Tunis
Software Engineer Intern Jul 2025 – Sep 2025
- Built a configurable RAG engine using Pinecone, reaching 94% semantic accuracy on the project evaluation benchmark; optimized embeddings to reduce latency by 45%.
- Developed a YAML-configurable chatbot platform enabling non-technical users to create 20+ bot variants; added automated tests that reduced the QA cycle by 50%.

Full-Stack Developer Mar 2025 – Jun 2025
- Built an AI-powered legal contract analysis platform for 40+ enterprise clients, processing 500+ reviews per month and reducing analysis time by 60%.
- Delivered React/Next.js interfaces, FastAPI REST APIs, and Docker-based CI/CD; reduced infrastructure setup time by 75%.

Frontend Developer Intern – Steros, Sfax Jul 2024 – Aug 2024
- Built a responsive company website and interactive UI components backed by PHP and MySQL.

Selected Projects

KubeMind – AI-Driven Kubernetes Incident Analyzer (3-person capstone) Source Code
- Built an AI-assisted Kubernetes incident-analysis workflow correlating events, logs, and metrics into structured reports using LangChain/LangGraph, Prometheus, Loki, Kafka, MongoDB, and FastAPI.

Multi-Tenant AI Chatbot Widget Source Code
- Developed an embeddable multi-tenant chatbot with Shadow DOM isolation and domain validation; integrated Hono, Supabase, and Upstash Redis and deployed on Railway/Cloudflare Pages.

PromptCraft – AI NFT Generation and Auction Platform Source Code
- Built an AI-generated artwork marketplace with real-time WebSocket/SSE bidding designed for 100+ concurrent users.

Technical Skills

Programming: Python, TypeScript, JavaScript, Java, C#, PHP, SQL
Software Engineering: Full-stack development, REST APIs, system design, automated testing, Git, Agile collaboration
Frontend / Backend: React, Next.js, Vue, FastAPI, Express.js, NestJS, Hono, ASP.NET Core
AI / Generative AI: LLM integration, RAG, embeddings, LangChain, LangGraph, prompt engineering
Data: MongoDB, MySQL, Supabase, Redis, Qdrant, Pinecone
Cloud / DevOps: Docker, Kubernetes, GitHub Actions, CI/CD, Prometheus, Loki, Kafka, Vercel, Railway, Cloudflare

Education

INSAT – National Institute of Applied Sciences and Technology Sep 2022 – Sep 2027 (expected)
Engineering Degree in Software Engineering (Master's-equivalent engineering degree)

Leadership & Activities

Event Mission Leader – Junior Enterprise INSAT 2024 – 2025
- Led a team of approximately 40 and the organizing committee for two flagship events, including Forum INSAT Entreprise (1,500+ attendees, 40+ companies).

Operations Manager – WinterCup 7.0 2025
- Coordinated logistics and on-site operations for an overnight coding contest with approximately 300 participants.